

Large-scale analysis of the integration of enhancer-enhancer signals by promoters

Reviewed Preprint

v2 • September 11, 2024

Revised by authors

Reviewed Preprint

v1 • November 3, 2023

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Abstract

Genes are often regulated by multiple enhancers. It is poorly understood how the individual enhancer activities are combined to control promoter activity. Anecdotal evidence has shown that enhancers can combine sub-additively, additively, synergistically, or redundantly. However, it is not clear which of these modes are more frequent in mammalian genomes. Here, we systematically tested how pairs of enhancers activate promoters using a three-way combinatorial reporter assay in mouse cells. By assaying about 69,000 enhancer-enhancer-promoter combinations we found that enhancer pairs generally combine near-additively. This behaviour was conserved across seven developmental promoters tested. Surprisingly, these promoters scale the enhancer signals in a non-linear manner that depends on promoter strength. A housekeeping promoter showed an overall different response to enhancer pairs, and a smaller dynamic range. Thus, our data indicate that enhancers mostly act additively, but promoters transform their collective effect non-linearly.

eLife assessment

Understanding how genomic regulatory elements control spatiotemporal gene expression is essential for explaining cell type diversification, function, and the impact of genetic variation on disease. This **important** study provides **solid** evidence that enhancers generally combine additively to influence gene expression. Moreover, promoters, particularly weaker ones, can exhibit supra-additivity when integrating enhancer effects. These findings highlight the context-dependent nature of enhancer-promoter interactions in gene regulation, and contribute to ongoing discussions about the selectivity and combination of regulatory elements.

<https://doi.org/10.7554/eLife.91994.2.sa3>

Introduction

Enhancers: background

Enhancers are DNA elements, typically a few hundred base pairs long, that bind transcription factors (TFs) and boost the activity of genes *in cis* (Banerji et al., 1981 [DOI](#)). Over the past decades, many insights have been obtained into the molecular architecture of enhancers, the roles of TFs and their cofactors, the interplay of these factors with chromatin, and the diversity of mechanisms that govern the interactions of enhancers with promoters (S. Kim & Wysocka, 2023 [DOI](#)).

MPRAs to probe enhancers

In recent years, massively parallel reporter assays (MPRAs) have been developed to systematically dissect the sequence features that determine the functionality of regulatory elements by probing thousands or even millions of DNA elements in a multiplexed manner (Inoue & Ahituv, 2015 [DOI](#); Klein et al., 2020 [DOI](#); Gallego Romero & Lea, 2023). Among others, two-way combinatorial MPRA studies that tested a large number of enhancer - promoter pairs revealed a variable degree of intrinsic compatibility between enhancers and promoters (Zabidi et al., 2015 [DOI](#); Bergman et al., 2022 [DOI](#); Martinez-Ara et al., 2022 [DOI](#)), in line with earlier evidence from analyses of single genomic loci (Bertolino & Singh, 2002 [DOI](#); Chang et al., 2004 [DOI](#); Vakoc et al., 2005 [DOI](#); Jing et al., 2008 [DOI](#); Deng et al., 2012 [DOI](#)).

Modes of enhancer interplay are still poorly understood

Surveys of mammalian genomes suggest that there are many more enhancers than genes (ENCODE Project Consortium, 2012 [DOI](#)). Indeed, genes are frequently controlled by multiple enhancers (Shlyueva et al., 2014 [DOI](#); Gasperini et al., 2020 [DOI](#); S. Kim & Wysocka, 2023 [DOI](#)). Often, such enhancers are clustered and jointly regulate a nearby tissue-specific gene (Tuan et al., 1985 [DOI](#); Grosveld et al., 1987 [DOI](#); Tolhuis et al., 2002 [DOI](#); Deng et al., 2012 [DOI](#); Hnisz et al., 2013 [DOI](#); Whyte et al., 2013 [DOI](#); Hay et al., 2016 [DOI](#); Schoenfelder & Fraser, 2019 [DOI](#); Brosh et al., 2023 [DOI](#)). However, how multiple enhancers act together to control a single promoter is still poorly understood. Several studies have found that enhancers can combine either additively (Bothma et al., 2015 [DOI](#); Lam et al., 2015 [DOI](#); Dukler et al., 2016 [DOI](#); Hay et al., 2016 [DOI](#); Will et al., 2017 [DOI](#); Lin et al., 2022 [DOI](#)), supra-additively (indicating synergy) (Bothma et al., 2015 [DOI](#); Lam et al., 2015 [DOI](#); Joo et al., 2016 [DOI](#); Shin et al., 2016 [DOI](#); Carleton et al., 2017 [DOI](#); Thomas et al., 2021 [DOI](#); Lin et al., 2022 [DOI](#); Brosh et al., 2023 [DOI](#)) or sub-additively (indicating redundancy) (Moorthy et al., 2017 [DOI](#); Osterwalder et al., 2018 [DOI](#)). Nevertheless, all of these studies tested only small numbers of enhancers in one or a few genomic loci. Therefore, it is still unclear what the globally predominant mode of functional interaction is between enhancers, and how this may depend on the target promoter. For this, more scalable approaches are needed. Here, we report such an approach.

Systematic approach

To systematically assess enhancer-enhancer interplay and the collective effect on promoter activity, we designed a three-way combinatorial MPRA based on our previous combinatorial reporters (Martinez-Ara et al., 2022 [DOI](#)). This assay allowed us to test tens of thousands of enhancer-enhancer-promoter combinations in a uniform setting (Inoue & Ahituv, 2015 [DOI](#); van Arensbergen et al., 2017 [DOI](#); Martinez-Ara et al., 2022 [DOI](#)). In order to test how different promoters might affect the enhancer interplay and their output, we not only varied enhancer pairs, but also the identity of the promoter in the three-way combinations.

Results and significance

By testing about 69,000 enhancer-enhancer-promoter combinations in mouse embryonic stem cells (mESCs) we found that pairs of enhancers mostly combine near-additively. Furthermore, promoter choice affects this enhancer-enhancer interplay, and promoters integrate the enhancer effects in a non-linear manner. Our three-way combinatorial approach provides further insights into two key aspects of gene regulation: how enhancers combine their effects, and how promoters integrate enhancer signals.

Results

Construction of libraries/experimental design

Triple combinations approach

To systematically test how pairs of enhancers work together and in turn activate promoters, we implemented a three-way combinatorial approach in a massively parallel reporter assay (MPRA). We designed a cloning strategy that enabled us to construct libraries of tens of thousands of reporters that each contain a different enhancer-enhancer-promoter (EEP) combination (**Figure 1A-B**).

Model system and promoter selection

For these experiments we chose mESCs as a model. We selected eight promoters that are active in mESCs. Of these, seven are from tightly regulated genes that are involved in the control of the pluripotency state of the cells or their exit from this state (*Klf2*, *Sox2*, *Nanog*, *Otx2*, *Lefty1*, *Fgf5*, *Tbx3*) (Acampora et al., 2013; Dunn et al., 2014; D.-K. Kim et al., 2014; Thomas et al., 2021). In addition, we included the promoter of the housekeeping (*Houkpe* et al., 2021) gene *Ap1m1*, which neighbours the *Klf2* gene. From each promoter we approximately included the -350- to +50-bp region around the transcription start site (TSS), because this includes the most information-rich part of functional promoters (Martinez-Ara et al., 2022; van Arensbergen et al., 2017).

Clusters of enhancers selection

We selected clusters of enhancers that are either known to regulate the above promoters (Blinka et al., 2016; Thomas et al., 2021; Zuin et al., 2022; Brosh et al., 2023), or are candidates for such function because of their proximity to the respective promoters (See methods and **Supplementary Table 1**) (Hnisz et al., 2013; Whyte et al., 2013). We also included putative enhancers from two additional developmental genes from the core pluripotency network (*Pou5f1* and *Nodal*) (Dunn et al., 2014) that are located elsewhere in the genome.

Definition and selection of single enhancers

We defined single enhancers as ~400 bp elements overlapping with DNase hypersensitive sites (DHSs) in clusters of enhancers (**Fig 1C-D**), as previously described (Martinez-Ara et al., 2022). Some of these enhancer clusters could be considered as single large enhancers (Long et al., 2016). However, since DHSs have been shown to retain enhancer activity individually (Barakat et al., 2018; Agrawal et al., 2021; Bergman et al., 2022; Martinez-Ara et al., 2022) we decided to test the combinatorial contributions of these smaller elements of an enhancer cluster. In total we selected 59 of such single enhancers. We also included 20 random sequences as negative controls (See Methods, **Supplementary Table 2**).

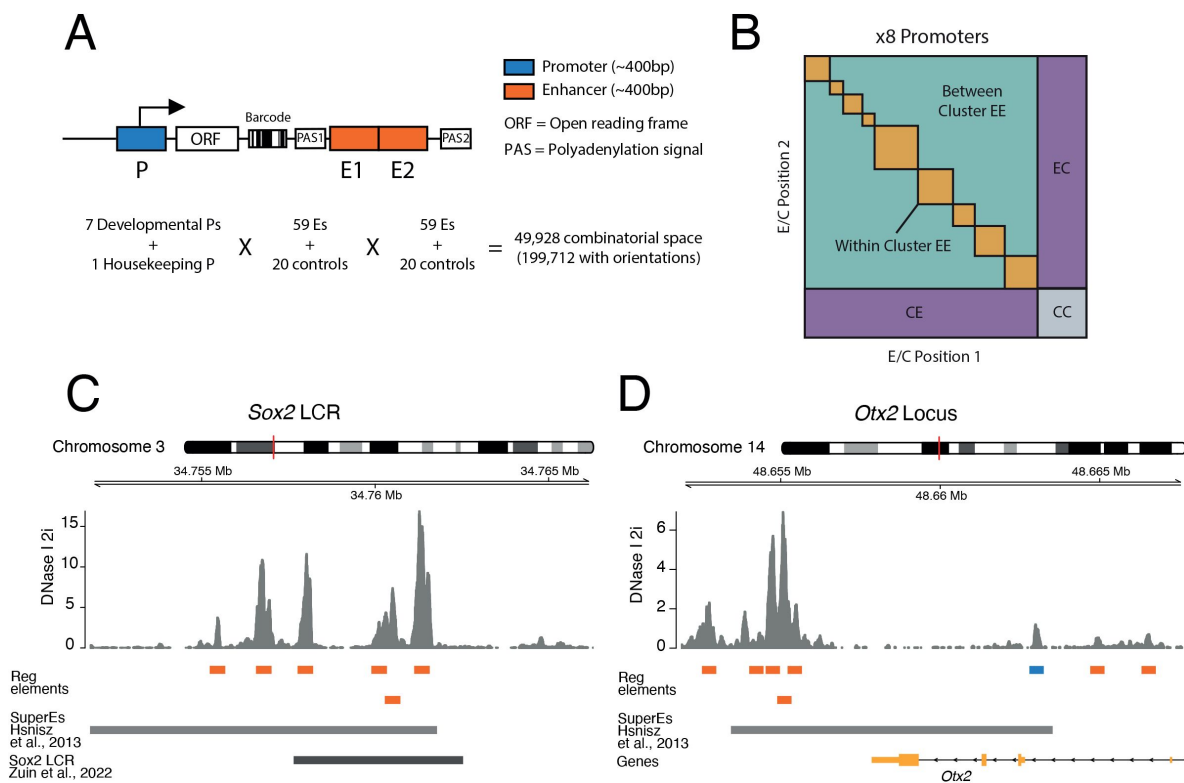


Figure 1.

Schematic of three-way combinatorial approach. **A**) Three-way combinatorial Massively parallel reporter assay (MPRA) design to test enhancer-enhancer-promoter combinations. 8 barcoded reporter assay libraries, 1 per promoter, were constructed. Pairs of DNA elements (enhancers and scrambled control sequences) were inserted after barcoded reporter. The enhancers and controls can be placed in both orientations in either the enhancer position 1 (E1) or enhancer position 2 (E2). **B**) The design of the library yields 8 matrices that contain control-control combinations (CC), enhancer-control combinations (EC and CE) and enhancer-enhancer combinations (EE). **C**) and **D**) 2 example loci, *Sox2* LCR (locus control region) (**C**) and *Otx2* (**D**) from where we selected enhancers to test in the reporter libraries. Enhancers (orange) were defined around DNase I hypersensitivity sites from mouse embryonic stem cells (mESC). Promoters (Blue) were chosen according to TSS annotation and mESC DNase I hypersensitivity sites. DNase I data is publicly available from [joshi et al., 2015](https://doi.org/10.1038/nature12874).

Cloning of libraries

We cloned each promoter separately into the reporter assay vector. We then barcoded the 8 separate vectors (see Methods). Next, we randomly combined enhancers and controls into pairs and cloned them into the eight different barcoded vectors downstream of the promoters (**Fig1A and Fig S1A**). This generated eight different EEP combinatorial libraries, one per promoter, with a total combinatorial space of 49,928 (8 promoters x 79 enhancers and controls x 79 enhancers and controls) or 199,712 if we consider the four possible orientation/position combinations of each enhancer or control fragment (**Fig1A**). Of this space, the libraries covered 138,528 (69%) of the total combinatorial space, and after application of stringent quality filters (see Methods) we were able to determine activities of 110,180 (55%) combinations (**Supplementary table 3**).

Explanation of library design

The design of the libraries generates eight matrices, one per promoter. These contain three types of combinations (**Figure 1B**): control-control (CC), enhancer-control (EC, regardless of position), and enhancer-enhancer (EE), which can be from the same or a different enhancer cluster. Control-control combinations are necessary to measure the baseline activity of each promoter. Enhancer-control combinations are used to estimate the effects of single enhancers. In total, we measured 2,584 control-control combinations, 38,390 enhancer-control combinations, and 69,206 enhancer-enhancer combinations.

Library transfections

We transfected each library separately into mESCs, with three biological replicates performed on different days. RNA was extracted and barcodes were counted by sequencing as previously described (Martinez-Ara et al., 2022) both from RNA and from the plasmid libraries. We calculated the activity of each barcode as the ratio between barcode counts in cDNA and plasmid DNA. We then averaged barcode activities for each triple combination across a minimum of 5 barcodes. For these averaged activities, all developmental promoter libraries showed high correlations between replicates (Pearson's $R = 0.82$ to 0.96) (**Figure S2**). The housekeeping promoter library showed somewhat lower correlations between replicates (Pearson's $R = 0.74$ to 0.78), which is most likely due to a smaller dynamic range. For further analyses, we then averaged the biological replicates for each promoter.

Effects of single enhancers

Calculation of boost indices

In order to quantify the activation by each enhancer we first calculated the baseline activity of each promoter, defined as the median activity across all control-control combinations. Then, for each individual enhancer-promoter combination, we calculated the median activity across all enhancer-control combinations. For each single enhancer we then calculated a boost index, which is the $\log_2$ ratio of its median activity across enhancer-control combinations over the promoter baseline across control-control combinations (**Figure 2A**). We observed that there was little position and orientation dependence of the enhancer effects (**Figure S3**). We therefore averaged the boost indices of single enhancers across positions and orientations for further analyses.

Selected DHSs are active enhancers

The single enhancer boost indices show that the majority of the sequences that we selected indeed act as enhancers, although the boost indices varied up to $\sim 5 \log_2$ units (**Figure 2A-B**). All tested enhancers showed significant activation of at least one promoter (at an estimated false discovery rate (FDR) ≤ 0.01) (**Figure 2B, D**). However, the dynamic ranges and effects of each enhancer

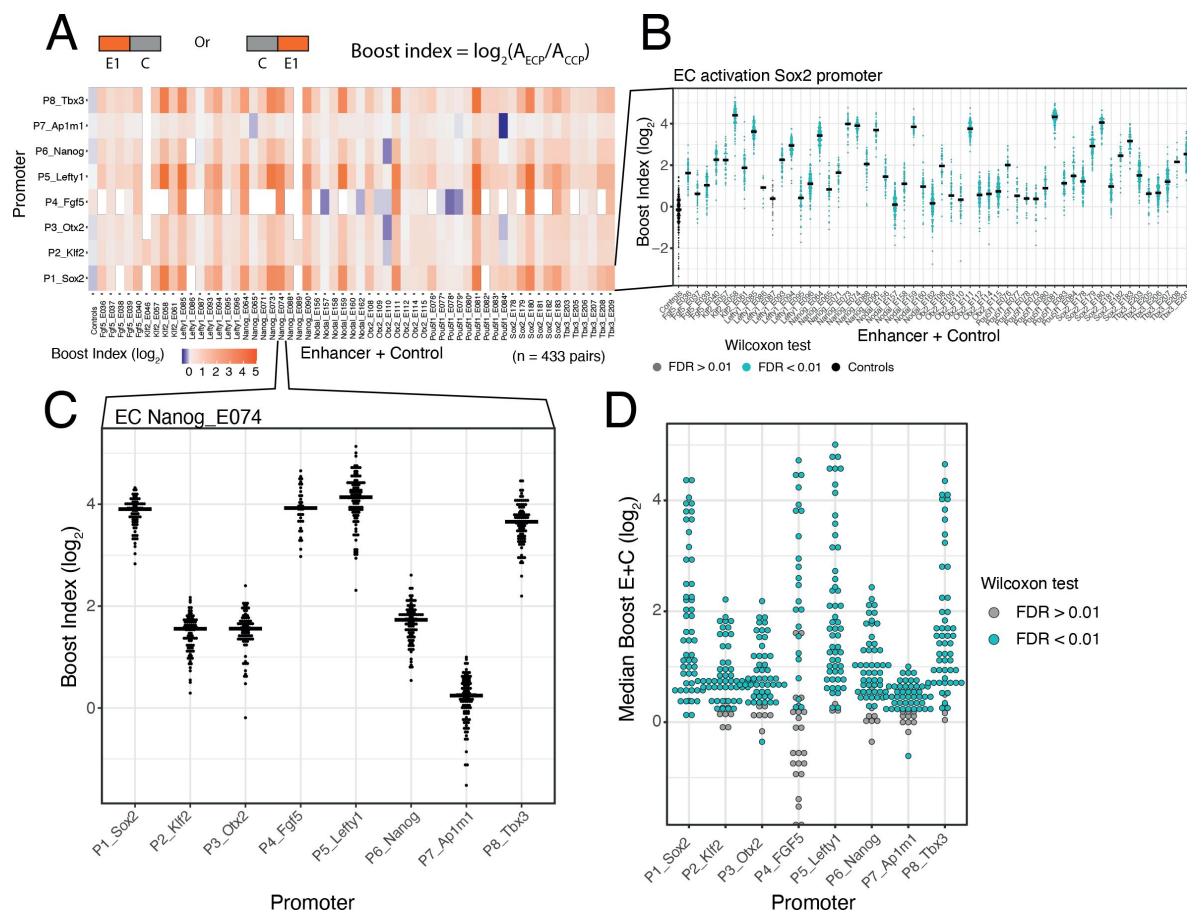


Figure 2.

Effects of single enhancers across promoters. **A**) Enhancer-promoter median boost index matrix of single enhancers. For each single enhancer a median boost index (see figure and methods, $\log_2(\text{Activity_Combination}/\text{Activity_control-control_baseline})$) was calculated across all enhancer-control combinations for that particular enhancer in any position and orientation, the baseline is the median activity of the promoter across all control-control combinations. For each control-control combination a boost index was calculated as the activity of the control-control combination over the median activity of all controls. A median control-control boost index was calculated based on this. Color coding of the matrix corresponds to the median boost indexes, white spaces are missing data. **B**) Distribution of boost indices for all enhancer+control combinations for the Sox2 promoter. Leftmost column corresponds to the boost index distribution for all control-control combinations for the Sox2 promoter. Each dot represents one enhancer+control combination. Horizontal lines correspond to the median of each distribution (same median as represented in the matrix in **A**). **C**) Distribution of boost indices for all enhancer+control combinations for the enhancer Nanog_E074 across all promoters. Each dot represents one enhancer+control combination. Horizontal lines correspond to the median of each distribution. **D**) Distribution of median boost indices for each single enhancer across promoters. Each dot represents the median of an enhancer across all enhancer+control combinations for that enhancers (same median as represented in the matrix in **A**). Coloring in **B** and **D** represents the significance at a 1% FDR for a Wilcoxon test comparing boost index distributions between enhancer+control combinations and the controls for each enhancer.

vary per promoter, with the boost indices being particularly smaller for the housekeeping promoter (*Ap1m1*) (Figure 2A, C). We conclude that most DHSs that we selected from the enhancer clusters act individually as enhancers.

Promoter selectivity of enhancers

In our previous study of a large set of E-P pairs we found that the majority of tested enhancers exhibited significant preference for certain promoters (Martinez-Ara et al., 2022). Applying the same statistical analysis to the current enhancer set confirmed this finding: the variability of enhancer effects across promoters are generally larger than what is expected from experimental noise (Figure S4). This indicates that the single enhancers tested here show significant promoter selectivity. This analysis (Figure S4) also points to systematic differences in promoter responses; this will be further highlighted below.

Enhancers generally combine near-additively

Enhancer-enhancer combinations are generally stronger than single enhancers

We then focused on the effects of enhancer-enhancer (EE) combinations. In the boost index matrix that covers all combinations for a single promoter, we observed that generally EE combinations were more active than EC combinations (Figure 3A). Indeed, quantitative analysis showed that EE combinations tend to induce stronger activation than the single enhancers (EC) (Figure 3B-C), although some saturation appears to occur with the strongest enhancers (Figure 3C).

Additive model fits better than multiplicative model

We then asked how the effects of two single enhancers are integrated in the enhancer-enhancer (EE) combinations. We considered two simple models: additivity and multiplicativity. For each EE combination we calculated the expected effect according to each of these models based on the single enhancer measurements, and then compared the observed and expected activities (additive in Figure 3D-E, multiplicative in Figure 3F, see Methods). Interestingly, for most of the promoters the additive model consistently showed a better fit to the measured data than the multiplicative model (Figure 3E-F, Figure S5). In particular, the multiplicative model tended to strongly overestimate activities at the higher range of activities (most active enhancers) for most promoters. When focusing on EE combinations for which the multiplicative and additive expected values differ more than $0.5 \log_2$ units (to limit random noise effects), we found that the multiplicative model matched the observed values better for only 0-17% EE combinations across the seven promoters. Only for the *Ap1m1* promoter the multiplicative and additive models were nearly indistinguishable. This may be due to the low dynamic range of activities observed with this promoter. We note that for the *Fgf5* promoter in the low activity ranges the expected activities were consistently underestimated for both models. It is likely that this is an inaccuracy due to the relative sparseness of the *Fgf5* data (the baseline activity of this promoter was estimated from only 23 measured control-control combinations, compared to 219-477 combinations for the other promoters). In summary, we conclude that a simple additive model fits the observed data better than a multiplicative model.

Supra-additive activation is rare and promoter dependent

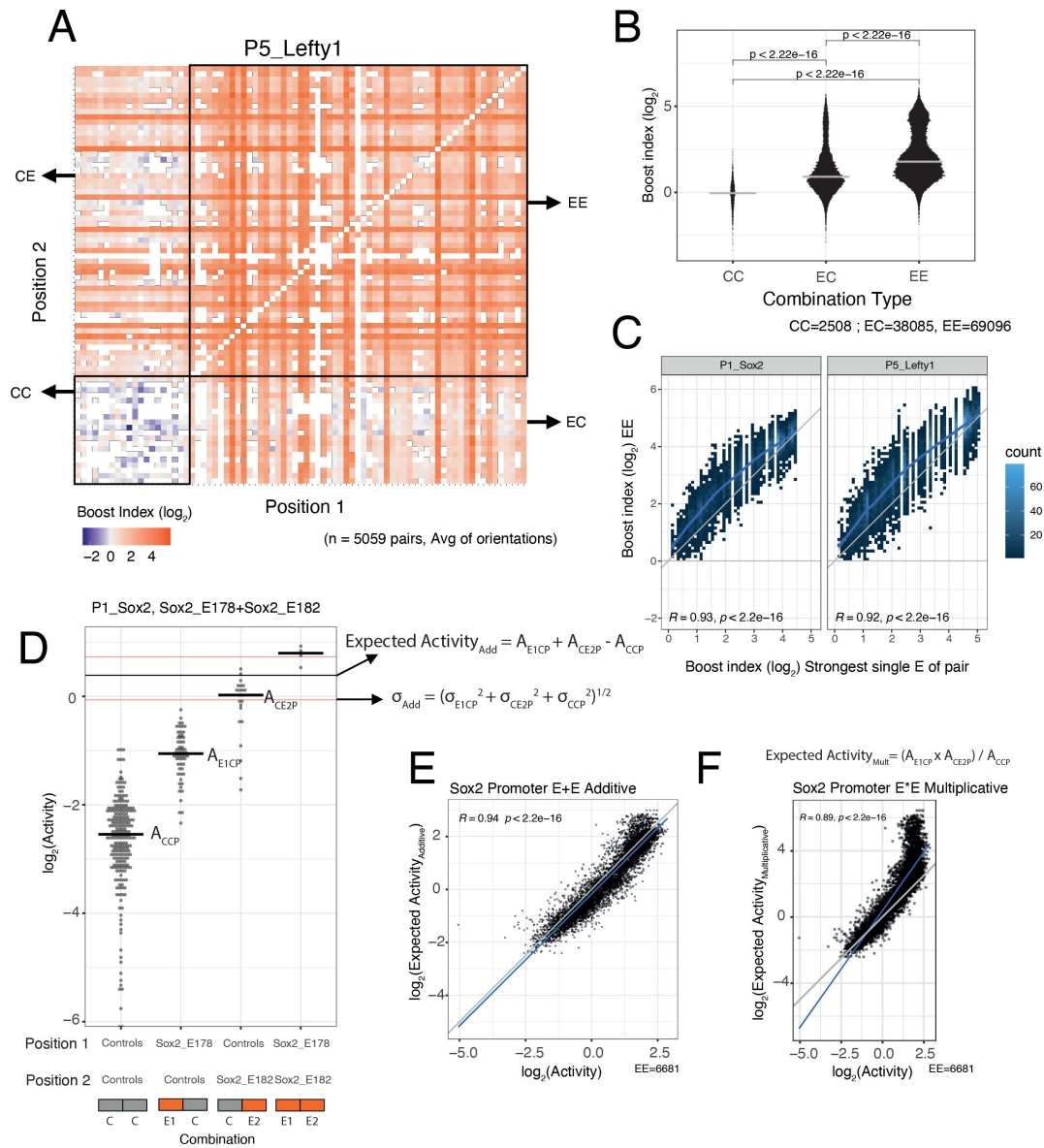


Figure 3.

Effects of enhancer-enhancer combinations. **A**) Fragment-fragment combinatorial boost index matrix for the *Lefty1* promoter. Each square represents 1 control-control (CC), enhancer-control (EC or CE) or enhancer-enhancer (EE) combination. Color coding corresponds to the average boost index for each combination across all orientations measured over the median control-control baseline. **B**) Boost index distributions across all 8 promoters for each combination type, control-control (CC), enhancer-control (EC regardless of position) or enhancer-enhancer (EE) combinations. P-values correspond to the result of a Wilcoxon test. **C**) Relationship between observed boost index for each EE combination and the observed boost index of the strongest single enhancer of the pair for *Sox2* and *Lefty1* promoters. Blue lines represent the LOESS fit of the data. **D**) Observed and expected additive activities for the *Sox2_E178+Sox2_E182* combination with the *Sox2* promoter and the individual activities of each of the elements. Each column represents the observed activities for the control-control combinations, the enhancer-control combinations and the enhancer-enhancer combinations. The horizontal black line represents the expected additive activity of the enhancer-enhancer combination as calculated by the formula in the panel in the linear space. The horizontal red lines represent the propagated standard deviations of the expected additive activity of the enhancer-enhancer combinations as calculated by the formula in the panel. **E and F**) Relationship between observed and expected activities (additive in **E**, multiplicative in **F**) for all enhancer-enhancer combinations for the *Sox2* promoter. The blue lines represent the linear fit of the data. Grey diagonal line is the x=y identity line. In all panels R represents Pearson's correlation. Expected activities are calculated in the linear space and then plotted in the $\log_2$ space.

Low frequency of supra-additive enhancer-enhancer interactions

Most enhancer-enhancer combinations show only minor deviations from the additive model (Figure S5 [↗](#)), as illustrated by the distributions of observed/expected ratios (Figure 4A [↗](#)). By applying a simple error model to estimate the noise in the expected EE activities according to the additive model (Figure 3D [↗](#), see Methods) we found that 74.5% of the measured EE activities lies within one standard deviation (SD) of the predicted activities (Figure 4A [↗](#)) and 94% lies within two standard deviations. This underscores that most of the tested enhancers combine near-additively, and that both sub-additive and supra-additive effects are generally rare or weak.

Supra-additivity occurs more frequently with weak promoters

Nevertheless, Figure 4A [↗](#) suggests that deviations from the additive model may vary between promoters. The *Sox2*, *Lefty1*, *Tbx3* and *Fgf5* promoters showed a relatively large proportion of EE pairs with supra-additive effects, while the *Ap1m1* promoter exhibited the lowest frequency of supra-additivity. Interestingly, the proportion of supra-additivity correlates inversely with promoter baseline expression (Figure 4B [↗](#), left panel). Thus, lowly active promoters allow more supra-additive EE interactions, although these remain a minority (<40% of EE pairs). In contrast, the proportion of sub-additive effects (i.e., >1 SD below expected activity) is very similar across promoters (Figure 4B [↗](#) right). This latter observation suggests that there is no strong saturation effect that could explain the decrease in number of supra-additive interactions as promoter activity increases.

EE supra-additive effects are relatively infrequent and often promoter-dependent

Next, we investigated whether supra-additivity involves a specific subset of enhancers. Visual inspection identified a few individual enhancers that exhibited supra-additive interactions with most other enhancers, but often in the context of particular promoters. An example is Nodal_E160 in combination with the *Fgf5* promoter (Figure 4D [↗](#), left panel). Other enhancers showed a diversity of supra-additive activities. Out of the 58 tested enhancers 32.7% (19/58) showed consistent supra-additivity in at least one promoter context ($>0.5 \log_2(\text{observed/expected})$, mean over all enhancers in at least one promoter context), and only 10.3% (6/58) showed consistent supra-additivity in two or more promoter contexts (Figure 4C [↗](#)). However, most enhancers behave like Sox2_E180 (Figure 4D [↗](#), right panel), showing small deviations from the additive behaviour with most other enhancers, and with all of the tested promoters (Figure 4C [↗](#)).

No evidence for locus-specific EE supra-additivity

We then asked whether enhancer pairs derived from the same genomic locus have a higher propensity towards supra-additivity than those not derived from the same locus. This could point towards co-evolution of enhancers that jointly control a gene. However, we found that EE combinations from the same locus showed on average not a higher observed/expected ratio (according to the additive model) than EE combinations from different loci; in fact, this ratio is even slightly lower (Figure 4E [↗](#)). Thus, generally there appears to be no preferential synergy among enhancers that are located in the same genomic locus.

Enhancer effects scale non-linearly across developmental promoters

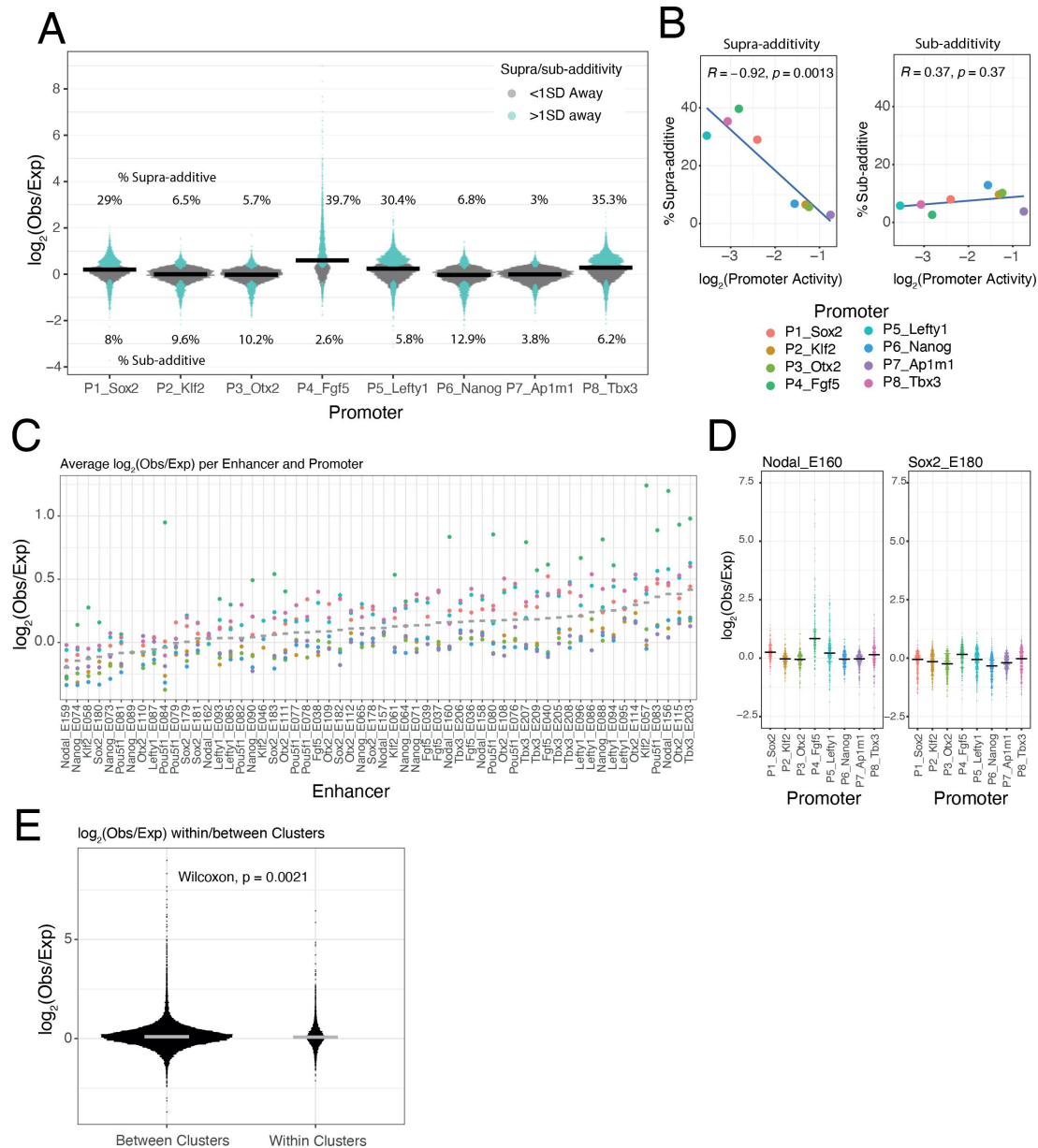


Figure 4.

Supra and sub-additive behaviours of enhancer combinations. **A**) Distributions of $\log_2$ observed activities over expected additive activities ratios of enhancer-enhancer combinations across promoters. Colored in turquoise are supra and sub-additive combinations for which the observed activity is more than 1 standard deviation away from the expected activity. Horizontal bars represent the median of each distribution. Numbers on the top part are the percentage of supra-additive combinations for each promoter. Numbers on the lower part are the percentage of sub-additive combinations for each promoter. **B**) Relationship between the percentages of supra and sub-additive enhancer-enhancer combinations and promoter control-control baselines. Blue lines are the linear fit of the data. R is Pearson's correlation. **C**) Average supra or sub-additive behaviour of each single enhancer across enhancer-enhancer combinations for each promoter. Each dot represents the median $\log_2$ observed over expected for all enhancer-enhancer combinations of a single enhancer and a particular promoter. Grey bars represent the median of each distribution. **D**) For 2 example enhancers, distribution of $\log_2$ observed over expected ratios for combinations of that enhancer with any other enhancer and promoter. Horizontal black bars represent the median of the distribution. **E**) Distribution of $\log_2$ observed over expected ratios for enhancer-enhancer pairs from the same enhancer cluster (within clusters) or from different enhancer clusters (between clusters). The p -value results from comparing both distributions using a Wilcoxon test. Horizontal grey bars represent the median of each distribution. In all panels Obs/exp refers to observed activity over expected additive activity.

Enhancer effects correlate well across developmental promoters

Prompted by our observations that some supra-additive EE interactions appear to be more pronounced in the context of specific promoters, we decided to systematically compare the boost indices of all EE pairs between the eight promoters. Strikingly, all seven developmental promoters showed strong pairwise correlations of their boost indices (Pearson's $R = 0.79$ to 0.98), whereas the housekeeping promoter (*Ap1m1*) correlated much less with each of the developmental promoters (Pearson's $R = 0.19$ to 0.60 ; [Figure S6](#)).

Slopes of promoter boost responses differ

Although the developmental promoters showed strong correlations, we were surprised to notice that the slopes of these relationships often deviated from 1. To further explore this, we plotted the boost indices of each promoter against the average boost indices across all eight promoters ([Figure 5A](#)). All of the developmental promoters correlated strongly with this average pattern (Pearson's $R = 0.93$ to 0.99) but this was not the case for the housekeeping promoter (Pearson's $R = 0.42$). The slopes relative to the averaged promoters ranged from 1.54 - 0.32 , i.e., about 5-fold, across the developmental promoters. These results imply that the developmental promoters differ in the scaling (i.e., steepness) of their responses to the set of EE pairs, even though the pattern of their responses is highly similar. Interestingly, the relative slopes negatively correlate with the baseline activity of these promoters ([Figure 5B](#)). Thus, promoters with low intrinsic activity respond more steeply to changes in combined EE activity than promoters with high intrinsic activity.

Promoter response differences also occur with single enhancers

Finally, we asked whether these differential responses of promoters require EE pairs, or can also be observed with single E elements. Indeed, this is the case ([Figure S7](#)).

Discussion

A large-scale survey of EEP combinations

Here, we employed a MPRA approach to measure three-way regulatory interactions in ~69,000 EEP combinations involving 8 promoters and 59 enhancers. This is of a much larger scale than previous efforts to elucidate the combinatorial interplay among enhancers, which typically focused on a handful of enhancers in single genomic loci ([Bothma et al., 2015](#); [Hay et al., 2016](#); [Shin et al., 2016](#); [Dukler et al., 2016](#); [Joo et al., 2016](#); [Lam et al., 2015](#); [Carleton et al., 2017](#); [Moorthy et al., 2017](#); [Osterwalder et al., 2018](#); [Thomas et al., 2021](#); [Lin et al., 2022](#); [Brosh et al., 2023](#)). These previous studies pointed to a diversity of functional enhancer-enhancer (EE) interactions, including redundancy, additivity and synergy. Our study also uncovered a broad spectrum of functional interactions, but simple near-additive effects of EE combinations were predominant, while supra-additive interactions occurred at a lower frequency and were dependent on the specific enhancer and promoter context.

Promoters “interpret” and scale EE interplay

An important finding is that promoters can have substantial impact on supra-additivity of enhancer-enhancer interactions. Promoters may thus be seen as “interpreters” of EE combinations. This underscores the importance of studying triple combinations of elements, and perhaps in the future even more complex settings. A salient finding is that the different promoters show different degrees of responsiveness to EE pairs, with the stronger promoters being boosted less than the weaker promoters. The responses of the promoters follow roughly a power-law function of which the exponent is dependent on the promoter ([Figures 5A](#) and [S6](#), which are on

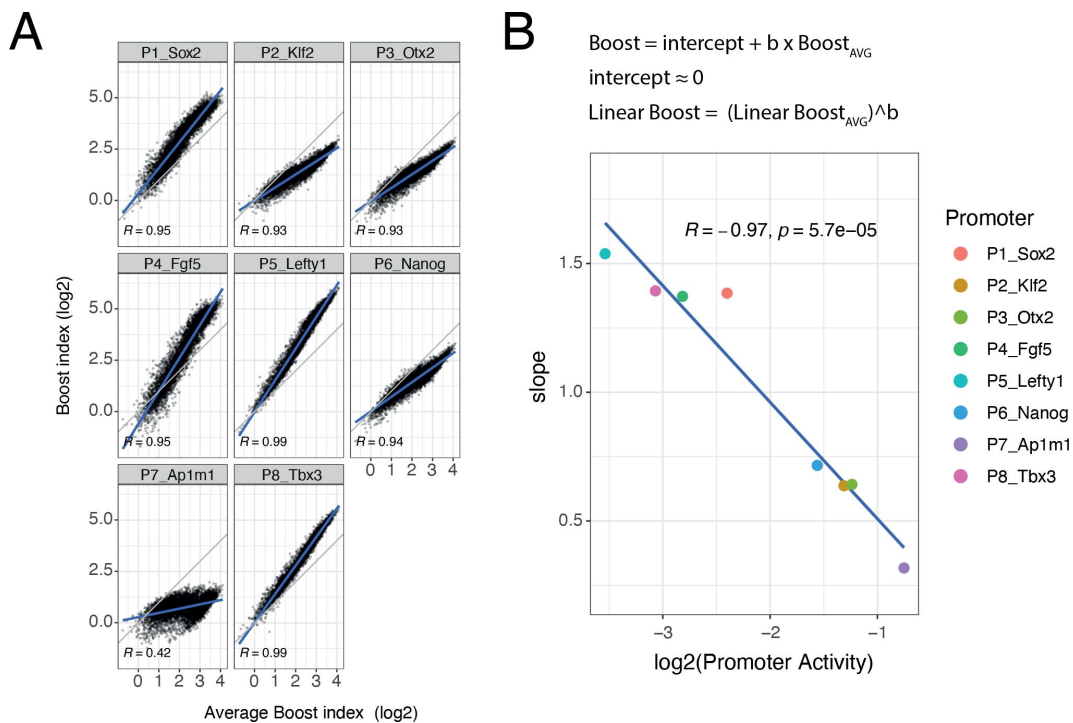


Figure 5.

Non-linear responses of promoters to enhancer-enhancer combinations. **A)** Relationship between observed boost indices and average boost index across promoters for all shared enhancer-enhancer combinations. Blue lines represent the linear fit of the data. **B)** Relationship between the slopes extracted from the linear fits in **A** and the baseline promoter activities derived from the control-control combinations. The formulae depict the relationship between the average boost indices and the observed boost indices of each promoter through the extracted slopes. For both panels R is Pearson's correlation.

log-log scale). Because it is not possible to define an absolute measure of enhancer strength, we cannot determine the absolute values of the exponents of the promoter power-law responses. This would be of interest because exponents <1 would lead to “diminishing returns” with higher EE activities, while exponents >1 would correspond to increasing non-linear amplification with higher EE activities. The apparent saturation observable at high enhancer activities in **Figure 3C** suggests that the exponents are most likely below 1. Our findings resemble those of an earlier study (Hong & Cohen, 2022) indicating that promoter activities are nonlinearly scaled by the genomic environment, with the steepness of the scaling being negatively correlated with promoter strength.

Episomal MPRA vs genomic context

The reporters in our MPRA were not integrated into the genome. On the one hand, this ensures that all EEP combinations are tested under similar conditions, without confounding effects of variable chromatin contexts. On the other hand, we cannot rule out that the results are not fully representative of a natural chromatin environment. However, other studies have indicated that transiently transfected reporters generally behave quite similar, although not identically, to integrated reporters (Inoue & Ahituv, 2015; Klein et al., 2020), and another study indicated that chromatin context can scale transcriptional activity of inserted reporters in a simple linear manner that is largely independent of the inserted sequence (Maricque et al., 2018). Some of the E elements that we tested were previously

Results may be biased by model system

We cannot rule out that our results are biased by our selection of enhancers and promoters, which was mostly focused on genes linked to pluripotency in mouse ESCs. We may have missed more prominent synergistic or other non-additive effects that may arise in different cell types or among different classes of enhancers and promoters. However, the protocol that we developed for three-way MPRA should be generally applicable to any set of enhancers and promoters.

Spacing and context of elements

It should also be considered that in our MPRA the enhancer and promoter elements are placed in fixed positions and close to one another, which does not recapitulate their relative positioning in the genome. Genomic distance and chromatin contexts could act as thresholds for activation, or turn additive interactions into non-linear relationships. For instance, the ability of a *Sox2* enhancer to activate its cognate promoter was found to depend on their distance, following a non-linear decay function (Zuin et al., 2022). The rules for EE interplay may depend on the EE distance or genomic context, as suggested by a recent report indicating that distant pairs of enhancers are more often synergistic than closely spaced enhancers, which tend to act additively (Lin et al., 2022). A recent study reported synergies between E elements in the *Sox2* Locus Control Region (a large compound enhancer region) in the native genomic context (Brosh et al., 2023). We did not observe such synergies, but the data of that study are not directly comparable to ours due to differences in the size of the elements tested. In the native *Fgf5* locus, several of the E elements exhibited some degree of synergy in differentiated cells (Thomas et al., 2021). However, in mESCs this synergy was not detected, which is in line our results for the same E elements from the *Fgf5* locus. Possibly, the default mode of enhancer-enhancer interplay is additive, while the precise chromatin context, element spacing and cellular state may enable synergy in some genomic loci.

Enhancer - promoter compatibility re-interpreted

In our previous MPRA study (Martinez-Ara et al., 2022) we found evidence for selectivity of enhancer - promoter functional interactions. However, due to the limited number of enhancer-promoter combinations tested we could not fully disentangle what caused these differences in promoter responses. The data from the eight promoters indicate that there are two intermingled

factors that account for differences between promoters. First, distinct classes of promoters show different specificities, as illustrated by the poor correlations between the *Ap1m1* housekeeping promoter and the developmental promoters. Second, promoters scale the input they receive from enhancer pairs differently, as discussed above. It is probable that the compatibility differences we observed in our previous study (Martinez-Ara et al., 2022) are a mix of these two phenomena: (i) promoters that respond differently to enhancers because of specificity, and (ii) promoters that respond differently to enhancers because of differences in responsiveness caused by their intrinsic strength.

Relevance of findings and context

The findings presented here offer insights into two aspects of the gene regulatory process – interplay between enhancers, and promoter integration of the resulting signals – in a more quantitative manner. The near-additivity of enhancers will help us understand how different enhancers may be more or less relevant in different loci. The non-linearity of promoter responsiveness and the effects of promoters on enhancer-enhancer interactions illustrates how some promoters may be more sensitive to activating signals. Thus, it will be useful to quantify promoter responsiveness genome-wide to understand which genes may be more or less sensitive than others to environmental cues or to non-coding mutations. These aspects may be especially relevant for developmentally regulated transcription factors and their target genes, like the ones here studied, as they can be very sensitive to dosage changes (Naqvi et al., 2023).

Methods

Cell culture

We used E14tg2a male mouse embryonic stem cells (ATCC CRL-1821) cultured in 2i+LIF for all experiments. As described {Martinez-Ara,2022,35594855}, we prepared 2i+LIF media according to the 4DN nucleome protocol (<https://data.4dnucleome.org/protocols/cb03c0c6-4ba6-4bbe-9210-c430ee4fdb2c/>).

The reagents used were Neurobasal medium (#21103-049, Gibco), DMEM-F12medium (#11320-033, Gibco), BSA (#15260-037; Gibco), N27 (#17504-044; Gibco), B2 (#17502-048; Gibco), LIF(#ESG1107; Sigma-Aldrich), CHIR-99021 (#HY-10182; MedChemExpress) and PD0325901 (#HY-10254; MedChemExpress), monothioglycerol(#M6145-25ML; Sigma) and glutamine (#25030-081, Gibco). Mycoplasma contamination was ruled out by monthly tests (#LT07-318; Lonza).

Selection of promoters, enhancers and controls

For the construction of the libraries we selected seven developmental genes from the mESC pluripotency regulatory network (Acampora et al., 2013; Dunn et al., 2014; D.-K. Kim et al., 2014; Thomas et al., 2021), *Klf2*, *Sox2*, *Nanog*, *Otx2*, *Lefty1*, *Fgf5*, *Tbx3*; and the housekeeping gene *Ap1m1* (Hounkpe et al., 2021), which is neighbouring the *Klf2* gene. The promoters of these genes were identified based on transcript and TSS annotation (Frankish et al., 2019) and overlaps with DNase hypersensitivity sites (DHS) (Joshi et al., 2015) as previously described (Martinez-Ara et al., 2022). We then selected sequences that approximately cover –350 to 50bp around the TSS.

The clusters of enhancers were selected in as follows: 1. Clusters of enhancers that were known to regulate the developmental genes (*Nanog*, *Sox2*, *Fgf5* clusters) (Blinka et al., 2016; Brosh et al., 2023; Thomas et al., 2021; Zuin et al., 2022); 2. Putative clusters of enhancers in proximity of one of the developmental genes and that overlapped with previously defined super-enhancers (*Lefty1*, *Tbx3*, *Otx2* clusters) (Hnisz et al., 2013; Whyte et al., 2013); 3. A cluster of putative enhancers previously identified by us (Martinez-Ara et al., 2022) to act as enhancers in an MPRA

(*Klf2* cluster). 4. Clusters of enhancers identified as putative enhancers of other mESC pluripotency genes for which we did not include their promoters (*Pou5f1* and *Nodal* clusters) and overlapping with previously defined super-enhancers (Hnisz et al., 2013 [DOI](#), 2015 [DOI](#); Whyte et al., 2013 [DOI](#)).

Clusters of enhancers were divided into single enhancers based on DNase hypersensitivity sites (DHS) (Joshi et al., 2015 [DOI](#)). These single enhancers were then resized to ~450bp from the center of DNase hypersensitivity peaks as previously described (Martinez-Ara et al., 2022 [DOI](#)).

For the library construction we included 20 450 bp long randomized control sequences as previously described (Martinez-Ara et al., 2022 [DOI](#)). Fifteen of these were based on five DHS peak sequences that were randomly scrambled. The other five controls were randomly generated sequences with the similar GC content as the DHS peak sequences. These control sequences were ordered as synthetic DNA. **Supplementary Table 2** contains the sequences of these controls.

For single enhancers and promoters we designed PCR primers against the 50 bp ends of each selected element using BatchPrimer3 v1.0 (You et al., 2008 [DOI](#)). This yielded PCR products of ~400 bp for each element. **Supplementary Table 1** lists the coordinates of all individual enhancers and promoters. **Supplementary Table 4** lists primer sequences.

EEP Library generation

The vector for library construction was previously used for the downstream reporter assay in our previous study (Martinez-Ara et al., 2022 [DOI](#)). This vector is based on a pSMART backbone. It contains a green fluorescent protein (GFP) open reading frame followed by a barcode, and a psiCheck polyadenylation signal (PAS) introduced during barcoding, followed by the cloning site for inserts and a triple polyadenylation site (SV40+bGH+psiCheckPAS).

Each of the selected eight promoters were amplified by PCR and individually inserted by Gibson assembly (#E2611S; New England Biolabs) into the reporter vector. Each construct was transformed into standard 5-alpha competent bacteria (#C2987; NEB) grown overnight in 500 ml of standard Luria Broth (LB) with 50 mg/ml of kanamycin and purified. Then each promoter sequence was verified by Sanger sequencing.

Each of the promoter vectors was then barcoded. Similar to what we described previously (Martinez-Ara et al., 2022 [DOI](#)), we digested 10 µg of each vector with AvrII (#ER1561; Thermo Fischer) and XcmI (#R0533; NEB) and performed a gel-purification. Barcodes were amplified in 10 separate 100ul PCR reactions using 5 µl of 10 mM primer 275JvA, 5 ml of 10 mM primer 465JvA and 1 µl of 0.1 mM template 274JvA (see **Supplementary Table 4** for sequences). 14 PCR cycles were performed using MyTaq Red Mix (#BIO-25043; Biotline), yielding 30 µg of barcodes. We purified barcodes by phenol-chloroform extraction and isopropanol precipitation. Barcodes were digested overnight with 80 units of NheI (#R0131S; NEB) and purified by magnetic bead purification (#CPCR-0050; CleanNA). Each promoter vector and barcodes were then ligated in one 100 µl reaction. In each reaction we used 3 µg of digested vector and 2.7 µg digested barcodes, 20 units NheI (#R0131S; NEB), 20 units AvrII, 10 ml of 103 CutSmart buffer, 10 µl of 10 mM ATP, 10 units T4 DNA ligase (#10799009001 Roche). We performed a cycle-ligation of six cycles (10 min at 22 °C and 10 min at 37 °C), followed by 20 min heat-inactivation at 80 °C. Linear barcoded vectors were purified by magnetic beads and digested for 3h with 40 units of XcmI (#R0533S; NEB). Finally barcoded vectors were purified again by gel-purification.

Single enhancers were amplified by PCR from E14tg2a mESC genomic DNA, then purified with magnetic beads (#CPCR-0050, Clean NA) and quantified by Qubit using dsDNA High sensitivity Qubit kit (#Q33231; invitrogen). Single enhancers and random controls were combined in approximately equimolar manner. This pool of elements was end-repaired using End-It DNA End-

Repair Kit (#ER0720; Epicentre) and self-ligated using Fast-link ligase (LK0750H; Lucigen). Duplets of 800bp-1000bp were excised from agarose gel, purified (BIO-52059; Bioline), and A-tailed using Klenow HC 3' → 5' exo (#M0212L; NEB).

The pool of duplets was ligated to each of the barcoded promoter vectors using Takara ligation kit version 2.1 (#6022; Takara). Ligation products were purified using magnetic bead purification and 2 µl of the ligation were electroporated into 20 µl of electrocompetent *E. coli* 10G supreme (#60081-1; Lucigen). To estimate the approximate complexity of the resulting libraries, dilutions of the electroporated bacteria were plated for overnight culture and colonies were counted. We aimed at a minimum of 2 million complexity per library in order to have sufficient expected representation of all possible combinations.

When the estimated complexity of the libraries was not high enough, we amplified the ligation products with primers aligning to the vector. This generated PCR products consisting of the duplets pool and a vector overhang. These PCR products were used for Gibson assembly with the barcoded promoters (#E2611S; NEB). Gibson assembly reactions were purified with magnetic beads and 2 µl of the reaction were electroporated into 20 µl of electrocompetent *E. coli* 10G supreme (#60081-1; Lucigen).

Each library was grown overnight in 500 ml of standard Luria Broth (LB) with 50 mg/ml of kanamycin and purified using a maxiprep kit (K210016, Invitrogen).

Characterization of libraries by iPCR and sequencing

Barcode to insert (enhancer and control duplets) combinations were identified by inverse PCR (iPCR) and Illumina sequencing as described before (van Arensbergen et al., 2017 [DOI](#)). In brief, each library was digested overnight with I-CeuI (#R0699S, NEB), barcode-insert fragments were then circularized, remaining linear fragments digested, and barcode-insert fragments were linearized again with I-SceI (#R0694S; NEB). Barcode-insert fragments were then amplified by PCR with Illumina adapters and sequenced on an Illumina NextSeq 550 platform using 150 bp paired-end sequencing. Each library was processed separately and mixed together for sequencing.

Transfection of libraries

Each EEP library was transfected separately into E14tg2a mESCs. Per library 20 million cells were nucleofected (5 µg of library and 5 million cells per cuvette, 5 cuvettes) using Amaxa nucleofector II, program A-30, and Mouse Embryonic Stem Cell Nucleofector™ Kit (#VPH-1001, Lonza). Three biological replicates were performed per library on different days. Cells were collected 24 hours after transfection in 5 ml of TRIsure (#BIO-38032; Bioline) and frozen at -80 °C until further processing.

cDNA sequencing

RNA was extracted and prepared for sequencing as described for the Downstream Assay in our previous study (Martinez-Ara et al., 2022 [DOI](#)). From each sample, the aqueous phase containing the total RNA of the TRIsure solution was extracted and purified on an RNA extraction column (#K0732, Thermo Scientific). Total RNA was digested with 10 units of DNase I (#04716728001; Roche) for 30 minutes in 10-8 10 µl reactions containing 5 µg of RNA. DNase I was inactivated by addition of 1 µl of 25 mM EDTA and incubation at 70°C for 10 min.

cDNA was produced in 20 µl reactions (1 per 10 µl DNase reaction) using Maxima Reverse transcriptase (#EP0743; ThermoFisher Scientific) and a gene specific (targeting the GFP reporter) primer (JvA304, see **Supplementary table 4** for sequence). dNTPs and the primer were mixed with the DNase digested RNA and incubated at 65°C for 5 minutes. The RT buffer, enzyme and RNase inhibitor were added and the reaction was incubated at 50 °C for 1 hour. cDNA was then

amplified by two nested PCRs to make it strand specific. The first PCR reaction uses (index variants of 285JvA containing the S2, index and p7 adaptor) and 305JvA (targeting the adapter introduced by 304JvA). Each 20 µl RT reaction was amplified in a 100µl PCR reaction with MyTaq Red mix (#BIO-25043; Biorline). The second PCR reaction was performed using 10 µl of the product of the previous reaction in a 100 µl reaction using the same index variant primer and index variants of 437JvA (containing the S1, index and p5 adaptor) (see **Supplementary table 4** for primer sequences). The first PCR was run for 10 cycles and the second one for 8 cycles using the recommended Mytaq Red mix conditions (for both PCRs: 1 min 96 °C, then each cycle 15 s 96 °C, 15 s 60 °C, 15 s 72 °C). PCR products were purified with magnetic beads and mixed for sequencing in an Illumina NovaSeq 6000 platform using 100bp single end reads.

Plasmid DNA (pDNA) sequencing

Plasmid libraries were processed for sequencing as described (Martinez-Ara et al., 2022 [↗](#)) but using dual indexing for sequencing on a NovaSeq 6000 platform with 100 bp single end reads. 1 µg of each plasmid was digested with I-SceI in order to linearise the plasmid. Barcodes were amplified by PCR for 9 cycles from 50 ng of material. Primers and reaction conditions were the same as in the amplification of cDNA. PCR products were purified using magnetic beads and mixed for sequencing.

Computational methods

cDNA and pDNA Data pre-processing and linking of barcodes to duplets was performed using a custom snakemake pipeline (Köster & Rahmann, 2012 [↗](#)). All other data analyses and quantifications were performed in R (R Core Team, 2022 [↗](#)). Figures were generated using ggplot2 (Wickham, 2016 [↗](#)). The snakemake pipeline and all the scripts used in this publication are available in a GitHub repository (<https://github.com/vansteensellab/EEPCombinations> [↗](#)).

Linking Barcodes to duplex inserts

The custom pipeline for linking barcodes to duplex inserts was previously described (Martinez-ara,2022, 35594855). In brief, iPCR reads from each library were locally aligned using bowtie (version 2.3.4) (Langmead & Salzberg, 2012 [↗](#)) with very sensitive parameters (–very-sensitive-local) on a custom bowtie genome. This custom Bowtie genome consists of virtual chromosomes that correspond to each of the enhancer and control sequences used to generate the combinatorial libraries. Bam files were processed by a custom Python script (Van Rossum & Drake, 2009 [↗](#)). This script extracts from read 1 the barcode sequence, and the identity and orientation of the DNA fragment in position 1. From read 2 it extracts the identity and orientation of the DNA fragment in position 2. Finally, barcodes were clustered using Starcode (version 1.1) (Zorita et al., 2015 [↗](#)) to remove PCR and sequencing errors.

cDNA and pDNA data pre-processing

For each cDNA and pDNA replicate of each library, barcodes were extracted from single-end reads using a custom Python script that matches the constant region after the barcode. Barcodes were clustered using Starcode (version 1.1) (Zorita et al., 2015 [↗](#)) to remove errors from sequencing and counts were summarized.

cDNA and pDNA data post-processing

For each library cDNA and pDNA barcodes were matched to iPCR barcodes. Any barcode assigned to multiple fragment identities was removed from the data. Per cDNA/pDNA replicate barcode counts were normalized to the total number of barcode counts. Then activity was calculated per barcode and cDNA replicate as the cDNA:pDNA normalized counts ratio. Per individual duplex

combination these normalized activity ratios were averaged across barcodes with a minimum requirement of five barcodes and eight pDNA counts per barcode. The mean activity across the three replicates was calculated as the geometric mean.

Calculation of boost indices

In our previous study (Martinez-Ara et al., 2022 [DOI](#)) we observed that it was critical to use control sequences to estimate the baseline activity of the promoters. For each promoter we used all of the control-control combinations to estimate its baseline activity. We calculated this as the median activity across all control-control combinations. Then the boost index of an enhancer-enhancer or enhancer-control (single enhancer boost index) combination was calculated as the $\log_2$ ratio between the observed activity of this combination and the baseline activity of the promoter. Except in **Supplementary Figure 3**, boost indices were averaged across positions and orientations. In **Figure 2a** [DOI](#) boost indices were averaged across all enhancer-control combinations for each enhancer. In **Figure 3a** [DOI](#) boost indices were averaged across orientations but not positions.

Selectivity of promoters

To analyse the selectivity of promoters on single enhancers (enhancer-control-promoter combinations) we used a Welch F-test due to the heteroscedasticity of the data. The test was performed on each of the populations of enhancer-control-promoter combinations. First the data was confirmed to be approximately normal by a Sapiro test, which confirmed that 94.1% of the 439 single enhancers + promoter combinations (that could be tested) show normal distributions (at a 1% FDR).

Calculation of additive and multiplicative expected activities

To calculate the expected additive and multiplicative activities we first had to determine three values. The baseline activity of the promoter (A_{CCP}) was measured as the median activity across all control-control combinations for each promoter. The single enhancer activity estimates (A_{E1CP} and A_{CE2P} , enhancer in position1 and position 2 respectively) were calculated as the median activity of all enhancer-control combinations where the enhancer was found in either position 1 (A_{E1CP}) or position 2 (A_{CE2P}). Therefore

$$ACCP = \text{median}\{AC1C2P, AC1C3P, \dots, ACiCjP\}$$

$$AE1CP = \text{median}\{AE1C1P, AE1C2P, \dots, AE1CiP\}$$

$$ACE2P = \text{median}\{AC1E2P, AC2E2P, \dots, ACiE2P\}$$

Then for each enhancer-enhancer-promoter combination the expected additive activity was calculated as the sum of the estimated activities of the single enhancers in each position minus the promoter baseline activity ($A_{E1CP} + A_{CE2P} - A_{CCP}$). The multiplicative expected activity was calculated as the product of the estimated activities of the single enhancers in each position divided by the promoter baseline activity ($(A_{E1CP} * A_{CE2P})/A_{CCP}$). For the expected additive activity, the standard deviation of the estimate was propagated as the square root of the sum of the variances of the single measurements. Therefore

$$\text{Expected Additive Activity} = (A_{E1CP} + A_{CE2P} - A_{CCP})$$

$$\text{Expected Multiplicative Activity} = (A_{E1CP} * A_{CE2P})/A_{CCP}$$

The additive expected activities were then used to calculate $\log_2$ (observed/expected) ratios in order to find supra-additive behaviours between enhancers. To find trends the $\log_2$ ratios were then averaged for each enhancer across enhancer-enhancer pairs.

Non-linear Promoter responses

For every enhancer-enhancer combination present in all eight libraries we averaged boost indices across the eight promoters. This average enhancer effect was then used to calculate the slope between the Average boost index and the observed boost index for each of the promoters. The slopes were calculated using the `lm` function in R.

Data availability

Raw sequencing data and processed data are available at Gene expression Omnibus (GEO): GSE240586. Publicly available datasets that were used are listed in and are available at <https://osf.io/yuc4e/>. Lab journal records are available at <https://osf.io/yuc4e/>. Code of data processing pipelines and analysis scripts are available at <https://github.com/vansteensellab/EEPCombinations>.

Author contributions

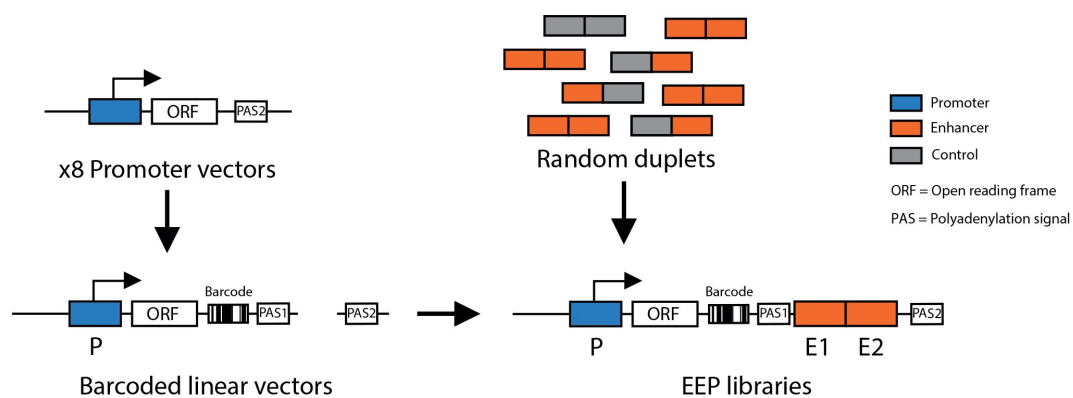
M.M.A., F.C., and B.v.S. designed the study. M.M.A. developed computational and experimental methods, performed analyses and experiments. B.v.S. and M.M.A. wrote the manuscript, with input from F.C. B.v.S. supervised the study.

Acknowledgements

We thank the NKI Genomics, and Research High Performing Computing facilities for technical support, and members of our laboratory for helpful discussions and comments. Funded by the European Union (European Research Council, RE_LOCATE, 101054449). Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. Research at the Netherlands Cancer Institute is supported by an institutional grant of the Dutch Cancer Society and of the Dutch Ministry of Health, Welfare and Sport. The Onco Institute is partially funded by the Dutch Cancer Society

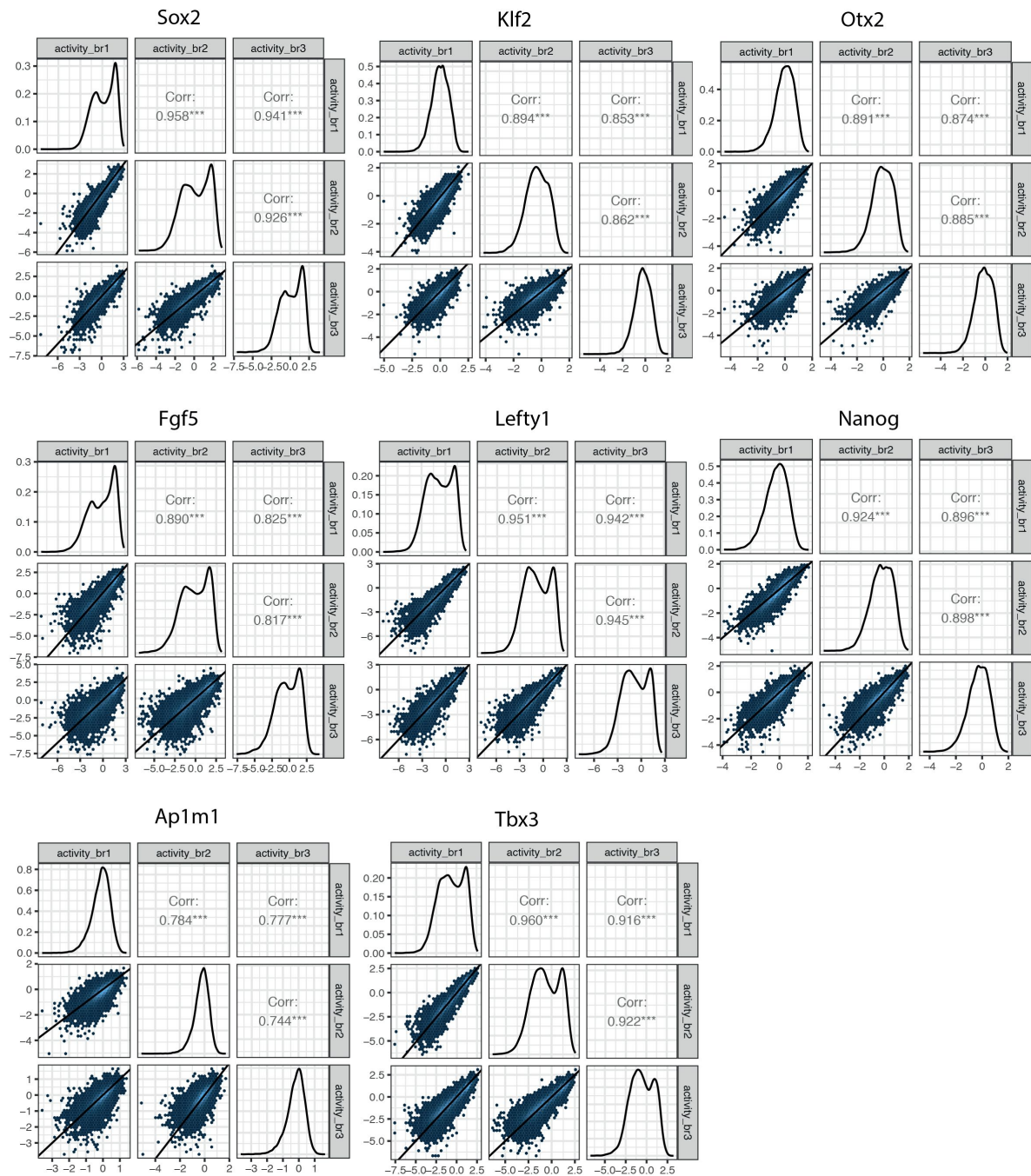
Declaration of interests

F.C. is a co-founder of enGene Statistics GmbH.



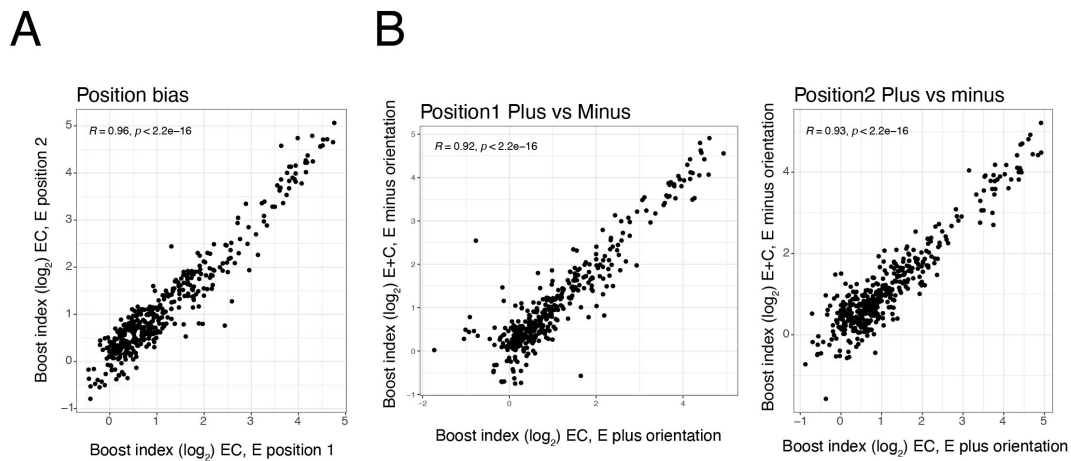
Supplementary Figure 1.

Schematic of cloning strategy to generate enhancer-enhancer-promoter libraries. First the 8 different promoters were cloned into a reporter vector. Each of the 8 vectors was barcoded and linearised, random enhancer-enhancer, enhancer-control and control-control combinations were cloned downstream of the barcoded reporter.



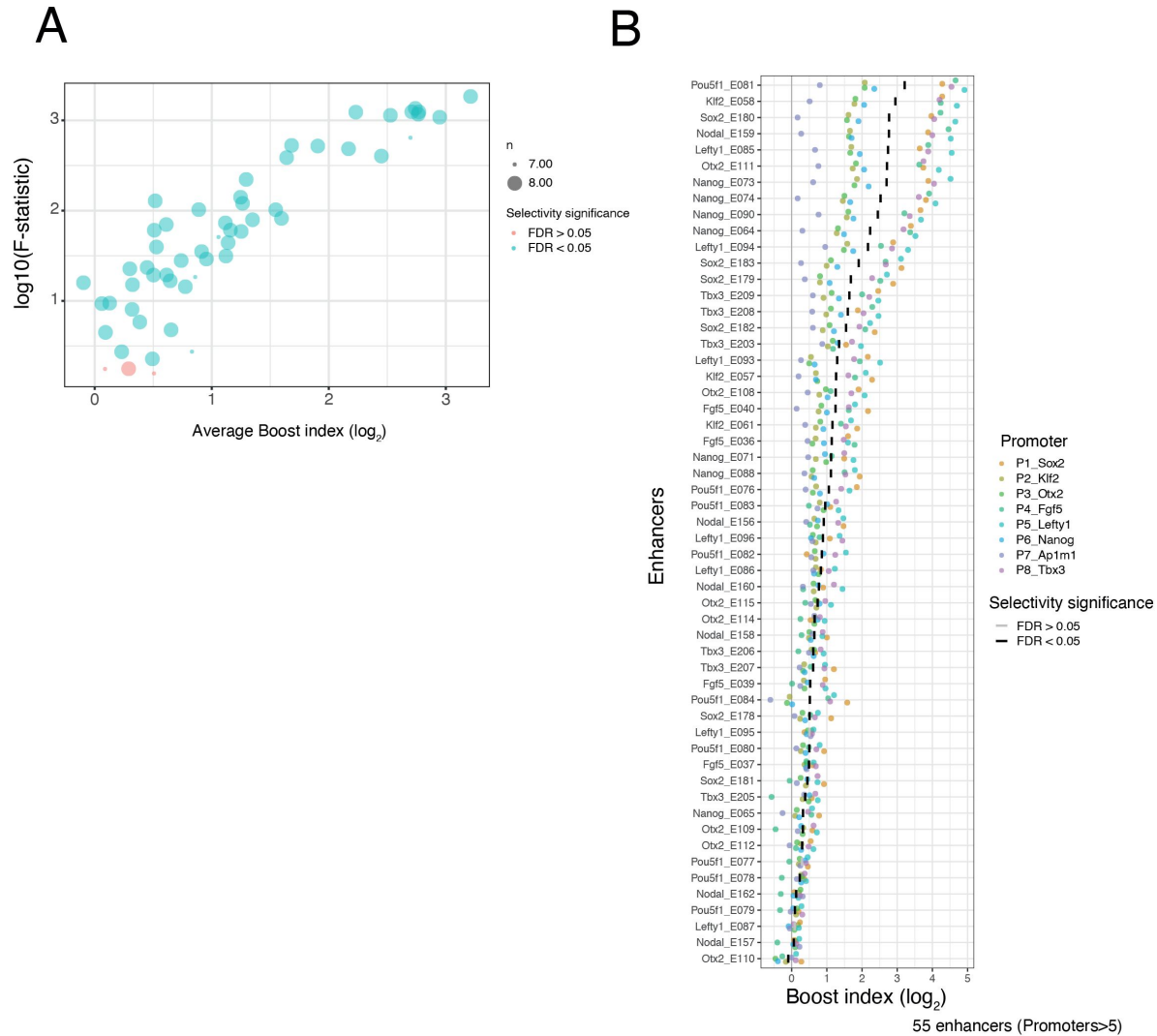
Supplementary Figure 2.

Reproducibility of experimental data. matrix of replicate correlations for all 8 EEP libraries. Lower left panels represent the 2 dimensional density plots of replicate-replicate activity correlations. Middle panels represent the 1 dimensional density plot of each replicate. Upper right panels are the Pearson's correlation of the mirror lower left panels.



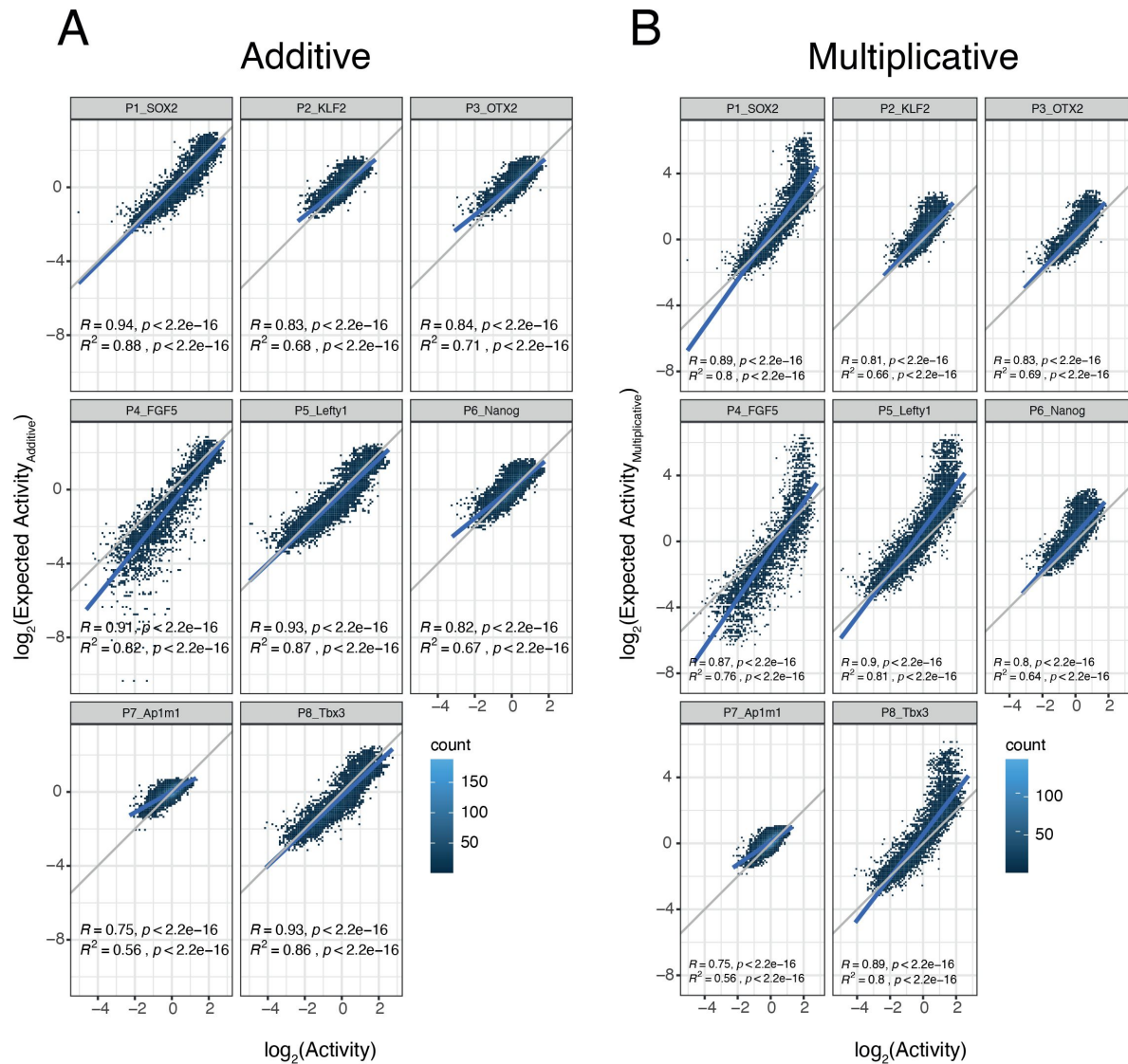
Supplementary Figure 3.

Position and orientation bias. **A)** Relationship between single enhancer boost indices across all EC combinations in position 1 versus in position 2 (See [Figure 1](#)). **B)** Relationship between single enhancer boost indices across all EC combinations plus versus minus orientation, in position 1(left) and in position 2 (right). For both panels each dot is one enhancer with one promoter. R is Pearson's correlation.



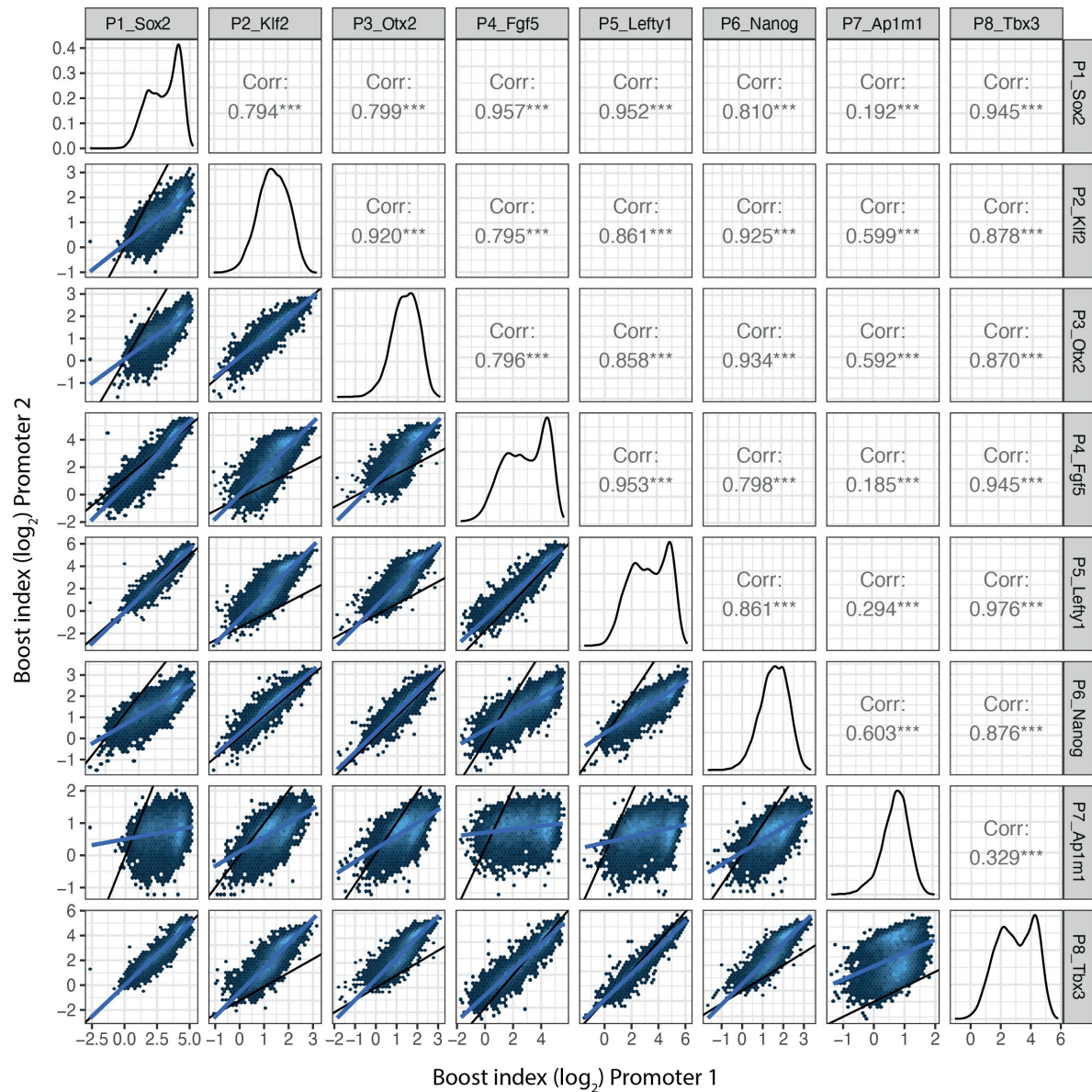
Supplementary Figure 4.

Selectivity of single enhancers. **A**) Relationship between average boost index of each single enhancer across all promoters and the F-statistic of a Welch F-test for that enhancer across all promoters. Colors indicate significance at a 5% FDR. **B**) Distributions of single enhancer boost indices for each enhancer across promoters. Each dot is one enhancer with one promoter and the boost index is the average boost index across all enhancer-control combinations. Colors of each point represents promoter identity. Vertical bars indicate the median boost index, and their grey or black shading indicates significance according to the Welch F-test at a 5% FDR.



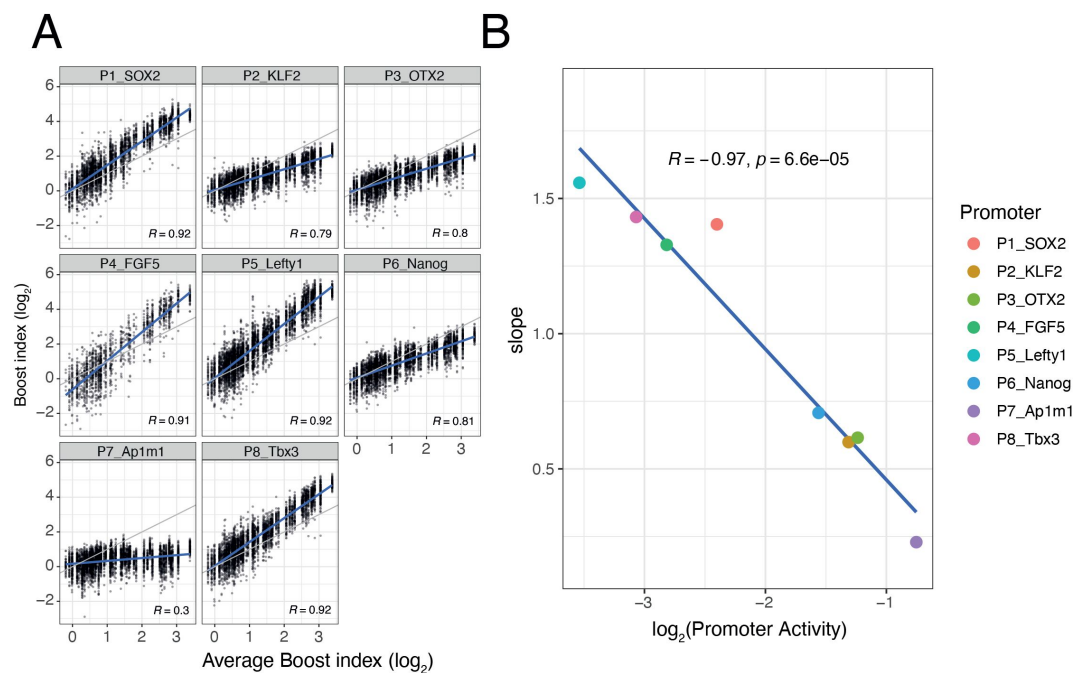
Supplementary Figure 5.

Additivity versus multiplicity for all promoters. Relationship between observed and expected activities (additive in **A**, multiplicative in **B**) for all enhancer-enhancer combinations for the all promoters. The blue lines represent the linear fit of the data. In all panels R and R^2 are based on Pearson's correlation.



Supplementary Figure 6.

Promoter promoter Boost index correlations for all shared enhancer-enhancer combinations. Lower left panels represent the 2 dimensional density plots of promoter-promoter boost index correlations. Middle panels represent the 1 dimensional density plot of each promoter. Upper right panels are the Pearson's correlation of the mirror lower left panels. Blue lines are the linear fit of the data.



Supplementary Figure 7.

(related to [figure 5](#)) Non-linear responses of promoters to enhancer-control (single enhancers) combinations. **A)** Relationship between observed boost indices of single enhancers and average boost index across promoters for all shared single enhancer-control combinations. Blue lines represent the linear fit of the data. **B)** Relationship between the slopes extracted from the linear fits in **A** and the baseline promoter activities derived from the control-control combinations. For both panels R is Pearson's correlation.

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Editors

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Reviewer #1 (Public review):

This manuscript by Martinez-Ara et al investigates how combinations of cis-regulatory elements combine to influence gene expression. Using a clever iteration on massively parallel reporter assays (MPRAs), the authors measure the combinatorial effects of pairs of enhancers on specific promoters. Specifically, they assayed the activity of 59x59 different enhancer-enhancer (E-E) combinations on 8 different promoters in mouse embryonic stem cells. The main claims of the paper are that E-E pairs combine nearly additively, and that supra-additive E-E pairs are rare and often promoter-dependent. The data in this study do generally support these claims.

This paper makes a good contribution to the ongoing discussions about the selectivity of gene regulatory elements. Recent works, such as those by Martinez-Ara et al. and Burgman et al., have indicated limited selectivity between E-P pairs on plasmid-based assays; this paper adds another layer to that by suggesting a similar lack of selectivity between E-E pairs.

An interesting result in this manuscript is the observation that weak promoters allow more supra-additive E-E interactions than strong promoters (Figure 4b). This nonlinear promoter response to enhancers aligns with the model previously proposed in Hong et al. (from my own group), which posited that core promoter activities are nonlinearly scaled by the genomic environment, and that (similar to the trend observed in Figure 5b) the steepness of the scaling is negatively correlated with promoter strength.

My only suggestion for the authors is that they include more plots showing how much the intrinsic strengths of the promoters and enhancers they are working with explain the trends in their data.

Specific Suggestions

Supplementary Figure 4 is presented as evidence for selectivity between single enhancers and promoters. Could the authors inspect the relationship between enhancer/promoter strength and this selectivity? Generating plots similar to Figure 4B and Figure 5B, but for single enhancers, should show if the ability of an enhancer to boost a promoter is inversely correlated to that promoter's intrinsic strength. Also, in Supplementary Figure 4, coloring

each point by promoter type would clarify if certain promoters (the weak ones) consistently show higher boost indices across all enhancers. If they do not, the authors may want to speculate how single enhancers can show selectivity for promoters while the effect of adding a second enhancer to an existing E-P has little selectivity. An alternate explanation, based solely on the strength of the elements, would be that when the expression of a gene is low the addition of enhancer(s) have large effects, but when the expression of a gene is high (closer to saturation) the addition of enhancer(s) have small effects.

Can anything more be said about the enhancers in E-E-P combinations that exhibit supra-additivity? Specifically, it would be interesting to know if certain enhancers, e.g. strong enhancers or enhancers with certain motifs, are more likely to show supra-additivity with a given promoter.

Comments on revised version:

The revised manuscript satisfactorily addresses the points I raised in the review. With the addition of the new graphs there is enough data for readers to decide whether the supra-additivity depends only on the strength of the promoter or on some other (undefined) feature of E-P pairs. This manuscript is a solid contribution to the ongoing debate about enhancer-promoter selectivity.

<https://doi.org/10.7554/eLife.91994.2.sa2>

Reviewer #2 (Public review):

Summary

This work investigates how multiple DNA elements combine to regulate gene expression. The authors use an episomal reporter assay which measures the transcriptional output of the reporter under the regulation of an enhancer-enhancer-promoter triple. The authors test all combinations of 8 promoters and 59 enhancers in this assay. There are two main findings: (1) enhancer pairs generally combine additively on reporter output (2) the extent to which enhancers increase reporter output over the promoter (individually and as enhancer-enhancer pairs) is inversely related to the intrinsic strength of the promoter. Both of these findings are interesting and are well supported by the data.

This study extends previous results on enhancer-promoter combinations to enhancer-enhancer-promoter triples. For example the near equivalence of Fig. 5b and Fig. S7b is intriguing. This experimental design also provides the ability to investigate the notion of selectivity (also commonly referred to as compatibility) between enhancer-enhancer pairs and promoters.

The authors note many limitations, including the selection of the elements and the size and spacing of the tested elements. Some of the enhancer-enhancer-promoter triples they test were also investigated by a different experimental design in Brosh et al 2023. Brosh et al observed non-additivity between these elements while this study did not. Ultimately we do not know which mechanisms produce the non-additivity that has been observed in native loci and which experimental designs would preserve such mechanisms.

Overall this is a nice experimental design and a great dataset for probing how enhancers and promoters combine to regulate gene expression. I have no major concerns, but I will try to clarify some methodological points I found confusing.

Methodology

The following two comments are meant to help the reader understand the methodology/terminology used in this paper and how it relates to other similar studies.

The interpretation that "promoters scale enhancer signals in a non-linear manner" is potentially confusing. I believe that the authors use "non-linear" to refer to the slopes (represented by the letter 'b' in Fig. 5b) being not equal to 1. Given how the boost index is defined, this implies the relationship

$$\text{Activity of EEP} = (\text{Activity of CCP}) * (\text{Average Linear Boost})^b$$

One potential source of confusion is that the Average Linear Boost term itself depends on the set of promoters that are assayed. Averaging across (many) promoters may alleviate this concern, in which case Average Linear Boost may be considered some form of intrinsic enhancer strength. If so, there is a correspondence between this terminology and the terminology presented in Bergman et al 2022. If b not equal to 1 refers to a non-linear scaling, then the reader may think that b=1 refers to a linear scaling. But if b=1, and the Average Linear Boost term is interpreted as intrinsic enhancer strength, then the equation above implies that the activity of EEP is equal to an intrinsic promoter strength times an intrinsic enhancer strength. This is essentially the relationship that is considered in Bergman et al 2022 and which is referred to in that paper as 'multiplicative'. The purpose of this comment is not to argue for what is the relationship that best explains the data, it is just to clarify the terminology.

Enhancer-promoter selectivity: As a follow-up to a previous study (Martinez-Ara et al, Molecular Cell 2022) the authors mention that the data in this study also shows that enhancers show selectivity for certain promoters. I found the methodology hard to follow, so this section of the review is meant to guide the reader in understanding how the authors define 'selectivity'. The authors consider an enhancer to be not selective if its 'boost index' is the same across a set of promoters. 'Boost index' is defined to be the ratio of the reporter output with the enhancer and promoter divided by the reporter output with just the promoter. Conceptually, I think that considering the boost index is a reasonable way to quantify selectivity. The authors use a frequentist approach to classify each enhancer as selective or not selective. The null hypothesis is that the boost index of the enhancer is equal across a set of promoters. This can be visualized in Fig. 2C where the null hypothesis is that the mean of each vertical distribution is equal. Note that in Figure S4b of this paper (and in Figure 4B of their 2022 paper) the within-group variance is not plotted. Statistical significance is assessed using a Welch F-test.

<https://doi.org/10.7554/eLife.91994.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

We thank the reviewer for the positive and constructive comments. We apologize for the very long delay in submitting this revised manuscript; due to personal circumstances we were not able to do this earlier.

This manuscript by Martinez-Ara et al investigates how combinations of cis-regulatory elements combine to influence gene expression. Using a clever iteration on massively parallel reporter assays (MPRAs), the authors measure the combinatorial effects of pairs of enhancers on specific promoters. Specifically, they assayed the activity of 59x59 different enhancer-enhancer (E-E) combinations on 8 different promoters in mouse

embryonic stem cells. The main claims of the paper are that E-E pairs combine nearly additively, and that supra-additive E-E pairs are rare and often promoter-dependent. The data in this study generally support these claims.

This paper makes a good contribution to the ongoing discussions about the selectivity of gene regulatory elements. Recent works, such as those by Martinez-Ara et al. and Burgman et al., have indicated limited selectivity between E-P pairs on plasmid-based assays; this paper adds another layer to that by suggesting a similar lack of selectivity between E-E pairs.

An interesting result in this manuscript is the observation that weak promoters allow more supra-additive E-E interactions than strong promoters (Figure 4b). This nonlinear promoter response to enhancers aligns with the model previously proposed in Hong et al. (from my own group), which posited that core promoter activities are nonlinearly scaled by the genomic environment, and that (similar to the trend observed in Figure 5b) the steepness of the scaling is negatively correlated with promoter strength.

We now discuss the parallel with the Hong 2022 study (Discussion, lines 307-310).

My only suggestion for the authors is that they include more plots showing how much the intrinsic strengths of the promoters and enhancers they are working with explain the trends in their data.

Agreed, see below.

Specific Suggestions

Supplementary Figure 4 is presented as evidence for selectivity between single enhancers and promoters. Could the authors inspect the relationship between enhancer/promoter strength and this selectivity? Generating plots similar to Figure 4B and Figure 5B, but for single enhancers, should show if the ability of an enhancer to boost a promoter is inversely correlated to that promoter's intrinsic strength...

Thank you for the suggestion, we have now repeated the analysis of Figure 5 for EP pairs instead of EEP triplets, and included it as new Supplementary Figure S7. Despite the lower statistical power, the trends are very similar.

...Also, in Supplementary Figure 4, coloring each point by promoter type would clarify if certain promoters (the weak ones) consistently show higher boost indices across all enhancers. If they do not, the authors may want to speculate how single enhancers can show selectivity for promoters while the effect of adding a second enhancer to an existing E-P has little selectivity. An alternate explanation, based solely on the strength of the elements, would be that when the expression of a gene is low the addition of enhancer(s) has large effects, but when the expression of a gene is high (closer to saturation) the addition of enhancer(s) have small effects.

We now added colour coding for each of the promoters in figure S4. We agree this clarifies the contribution of each promoter to the selectivity of each enhancer and it further confirms the responsiveness trends observed in Figure 5.

Can anything more be said about the enhancers in E-E-P combinations that exhibit supra-additivity? Specifically, it would be interesting to know if certain enhancers, e.g. strong enhancers or enhancers with certain motifs, are more likely to show supra-additivity with a given promoter.

Unfortunately, even with the number of enhancers that we tested, we lack statistical power to identify sequence motifs that may favour supra-additivity.

Reviewer #2 (Public Review):

We thank the reviewer for the supportive and constructive comments. We apologize for the very long delay in submitting this revised manuscript; due to personal circumstances we were not able to do this earlier.

Summary

This work investigates how multiple regulatory elements combine to regulate gene expression. The authors use an episomal reporter assay which measures the transcriptional output of the reporter under the regulation of an enhancer-enhancer-promoter triple. The authors test all combinations of 8 promoters and 59 enhancers in this assay. The main finding is that enhancer pairs generally combine additively on reporter output. The authors also find that the extent to which enhancers increase reporter output is inversely related to the intrinsic strength of the promoter.

This manuscript presents a compact experiment that investigates an important open question in gene regulation. The results and data will be of interest to researchers studying enhancers. Given that my expertise is in modeling and computation, I will take the experimental results at face value and focus my review on the interpretation of the results and the computational methodology. I find the result of additivity between enhancers to be well supported. The findings on differential responsiveness between promoters are very interesting but the interpretation of such responses as 'non-linear' or 'following a power-law' may be misleading. More broadly, I think a more rigorous description of the mathematical methodology would increase the clarity and accessibility of this manuscript. A major unanswered question is whether the findings in this study apply to enhancers in their native genomic context. Regardless, investigating such questions in an episomal reporter assay is valuable.

Main comments

Applicability to native genomic context: The applicability of the results in this paper to enhancers in their native genomic context is unclear. As the authors state in the discussion section, the reporter gene is not integrated into the genome, the spacing between enhancers does not match their native context etc. It is thus unclear whether this experimental design is able to detect the non-additivity between enhancers which is known to be present in the genome. This could be investigated by testing the enhancer-enhancer-promoter tuples for which non-additivity has been observed in the genome (references are given in the introduction) in this assay.

We appreciate the suggestion, but we chose not to go back to the lab to generate additional data to address this point. Of the cited previous studies, two are comparable to our study because they also used mESCs and included loci that we also studied: Thomas et al. (2021) and Brosh et al. (2023). We now discuss how the findings of these two studies relate to our observations in the Discussion, lines 336-345.

Interpretation of promoter responses as non-linear and following a power-law: In Fig 5, the authors demonstrate that enhancer-enhancer pairs boost reporter output more for weak promoters as opposed to strong promoters. I agree the data supports this finding, but I find the interpretation of such data as promoters scaling enhancers according to a power-law (as stated in the abstract) to be misleading. As mentioned on line 297, it is not possible to define an intrinsic measure of enhancer strength, thus the authors assign the

base of the power-law to be the average boost index of the enhancer-enhancer pair across the 8 promoters. But this measure incorporates some aspect of a promoter and is not solely a property of enhancers...

We agree that the power-law conclusion in the abstract was too strong; we have rephrased it as "non-linear".

...It would also be useful to know whether the results in Fig 5 apply to only enhancer-enhancer-promoter triples or also to enhancer-promoter pairs.

We have now added this EP analysis as new Supplemental Figure S7. Although the statistical power is much lower, this shows very similar trends as the EEP analysis. We briefly report this, lines 275-278.

Enhancer-promoter selectivity: As a follow-up to a previous study (Martinez-Ara et al, Molecular Cell 2022) the authors mention that the data in this study also shows that enhancers show selectivity for certain promoters. The authors mention that both studies use the same statistical methodology and the data in this study is consistent with the data from the 2022 paper. However, I think the statistical methodology in both studies needs further exposition. This section of the review is thus meant to ensure that I understand the author's methodology, to guide the reader in understanding how the authors define 'selectivity', and to probe certain assumptions underlying the methodology.

My understanding of the approach is as follows: The authors consider an enhancer to be not selective if its 'boost index' is the same across a set of promoters. 'Boost index' is defined to be the ratio of the reporter output with the enhancer and promoter divided by the reporter output with just the promoter. Conceptually, I think that considering the boost index is a reasonable way to quantify selectivity.

The authors use a frequentist approach to classify each enhancer as selective or not selective. The null hypothesis is that the boost index of the enhancer is equal across a set of promoters. This can be visualized in Fig. 2C where the null hypothesis is that the mean of each vertical distribution is equal. Note that in Figure S4 of this paper (and in Figure 4B of their 2022 paper) the within-group variance is not plotted. Statistical significance is assessed using a Welch F-test. This is a parametric test that assumes that the observations within each vertical distribution in Fig 2C are normally distributed (this test does allow for heteroskedasticity - which means that the variance may differ within each vertical distribution). Does the normality assumption hold? This analysis should be reported. If this assumption does not hold, is the Welch test well calibrated?

We have tested the normality of all of the single enhancer + promoter combinations that were tested using the welch F-test. 94.1% of the 439 single enhancers + Promoter combinations show normal distributions (at a 1% FDR). We have added this to the methods section of the revised manuscript. Apart from this, non-normality has little to no influence on the Welch F-test performance (<https://rips-irsp.com/articles/10.5334/irsp.198>). Therefore, the use of the Welch F-test to score enhancer selectivity on these data is valid. Apart from this, we agree that a simple binary classification of selective vs non-selective is not descriptive enough for these kinds of data. We addressed this in our previous publication by exploring the relationship between selectivity and enhancer strength. However, in the objective in this publication was solely to show that this new dataset follows similar selectivity patterns to our previous publication. Furthermore, our analysis on the non-linearity of promoter response is a more quantitative continuation on the analysis on selectivity as this is probably one of the major contributors to enhancer selectivity. This was probably present in our previous paper but could not be analyzed as there were less combinations per promoter.

For further clarity, we have now highlighted the individual promoters in Figure S4 by colors.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

I found this to be an interesting manuscript and am glad this experiment was conducted. As I wrote in my public review, I think that clarifying the computational methods/ideas would really help. I also think it would be helpful to properly define the terms that are being used. For example, this manuscript uses the terminology cooperativity and synergy. Are these meant to be synonymous with supra-additivity?

Thank you for this point. The revised manuscript no longer uses the word “cooperativity”. We now use “supra-additivity” when describing our data, and “synergy” as biological interpretation. In the Introduction we now clarify this distinction.

Comments on enhancer selectivity:

In the public review, I have given comments on the statistical methodology employed to assess enhancer selectivity. On a more subjective note, I'm not convinced that a frequentist approach to a binary classification of 'selective' vs 'not selective' is that useful here. I think it would be more useful to report an 'effect size' of the extent to which an enhancer is selective and to study the sources of this effect size. I think you've tried to do this in lines 329-339 of the discussion but I think the exposition could be clearer.

Figure S4B may suggest how to do this. It appears that the distribution of boost indices for a given enhancer is trimodal (this is most obvious for the stronger enhancers on the top of the plot). Is it the case that each mode (for each enhancer) consists of the same set of promoters? I think what is implied by Figure 5B is that the stronger promoters are not boosted as much as the weaker promoters. So does the leftmost mode consist of Ap1m1, the middle mode consist of Klf2/Otx2/Nanog, and the rightmost mode of Sox2/Fgf5/Lefty1/Tbx3? If so, I would recommend emphasizing this in the text/figure and clarifying how this relates to selectivity. It seems that the chain of logic is as follows: (1) We define an enhancer to be selective if its boost indices across a set of promoters are not the same. (2) We generally observe that stronger promoters get boosted less than weaker promoters. (3) Thus selectivity arises due to differences in intrinsic strengths of the promoter. I think this is what is being implied in lines 329-339 of the discussion, but it took me multiple readings to understand this and I'm not convinced the power-law explanation is justified (see public review).

We have modified this paragraph of the Discussion (now lines 350-359).

Regarding the power-law: in the Results we state “roughly a power-law function”. We have removed the power-law claim from the abstract, that conclusion as phrased was indeed too firm.

Reference to Zuin et al

Lines 323 - 325: A reference is made to the data from Zuin et al "following approximately a power-law". What data in Zuin et al does this statement refer to? I do not believe the authors in Zuin et al claim that the relationship between GFP intensity and enhancer-promoter distance (Figure 1h,i from Zuin et al) follows a power law. It is certainly non-linear, but I have taken a look at this data myself and do not find it follows a power-law. Please either explain this further and rigorously justify the claim or adjust the wording accordingly.

Good point, in the discussion of Zuin et al we have replaced “power law” with “non-linear decay function”

<https://doi.org/10.7554/eLife.91994.2.sa0>